CONCEPT PAPER

Our Idea

RideAware's intelligent cycling glasses turn safety from reactive to proactive. The wearable assistance system uses sensor fusion, GPS and an AR display module to project important real-time alerts directly into the cyclist's field of vision.

NEW ENTERPRISE DEVELOPMENT PROJECT

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Date of Submission	13.10.2025

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LIST OF ABBRIVIATIONS

AR……...……Augmented Reality

DAS……Driving Assistance System

IMU……Inertial Measurement Unit

MCU……Microcontroller Unit

BLE…….. Bluetooth Low Energy

ANT+ ……..……....Advanced and Adaptive Network Technology

VIO……..... Visual Inertial Odometry

SLAM……..…….……Simultaneous Localization and Mapping

AR HUD …….……Augmented Reality Heads Up Display

TTC……. ……..Time to Collision

TTO……. ……..Time to Overtake

TIR…….…. …….....Total Internal Reflection

LCoS …….....Liquid-crystal-on-silicon

GNSS……. ……...Global Navigation Satellite System

EXECUTIVE SUMMARY

I. Introduction

The introduction and background portion of the report details various cyclist statistics and data trends collected from various sources, globally and throughout Ireland, to demonstrate the overall need for greater cyclist accident prevention and road safety.

II. Our Idea

RideAware's intelligent cycling glasses turn safety from reactive to proactive. The wearable assistance system uses sensor fusion, GPS and an AR display module to project important real-time alerts directly into the cyclist's field of vision.

III. Ideation Process

We approached the ideation process with the Blue Ocean Strategy in mind to tap into a new market. Each team member developed initial concepts independently, followed by a reflection and group to ensure unbiased selection. Through critical questioning, diverse perspectives, and expert feedback, we refined our ideas into a unique, user-centred innovation.

V. Technical Evaluation

RideAware frames integrate a global-shutter monocular camera, 6-axis IMU, rear radar and GNSS. VIO/SLAM fuses streams; the MCU computes TTC/TTO hazards, then renders world-locked AR via waveguide HUD plus directional bone-conduction audio. BLE links radar, glasses and app. Heavy vision processing is offloaded to the smartphone for battery and thermal efficiency.

IV. Market Analysis

Cycling in Dublin is underutilised due to safety concerns, poor infrastructure, and traffic risks. Bicycles are increasingly seen as practical for short- to medium-length trips and offer significant health, convenience, and environmental benefits. Solutions that assist in improving safety and confidence represent a clear market opportunity.

VI. Investments & Return

We took on the investment and returns segment with a scenario analysis approach, giving estimates of sales revenues using a 'best-case', base-case' and 'worst-case' approach. This was supplemented by a discrete exponential growth regression model to highlight the exponential growth nature often seen in tech startups.

See Beyond

Mission

To transform urban cycling by providing smart, hands-free solutions that improve safety, confidence, and mobility for all riders.

TEAM DETAILS

Group 35 – RideAware Ltd.

Juliana Dury
CEO

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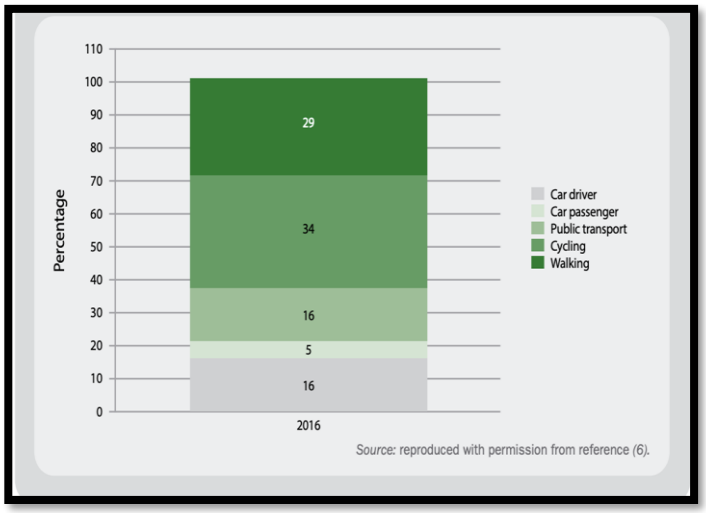
1. INTRODUCTION

On the 29th of August 2024, two brothers Johnny and Matthew Gaudreau, who were heavily recognised as icons across the hockey community were struck down and killed by a drunk driver while cycling in the off-season (USA Hockey, 2024). This unfortunate tragedy heavily resonates with members of our team, which naturally contributed to the development of RideAware.

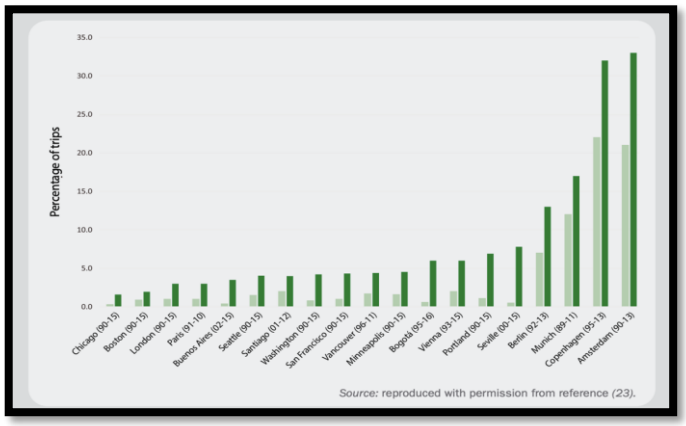
As mentioned, even amongst icons, bicycle accidents and deaths are a common occurrence and a growing fear across the world. However, despite this risk, many people continue to opt for cycling over automobiles and buses due to the obvious health and environmental benefits, such as its convenience in physical exercise and the overall reduction of greenhouse gases emitted. The following chart taken from a recent study by the World Health Organization (2020, p. 3), demonstrates the proportion of people who chose cycling as their primary mode of transportation, in a selected European country:

Exhibit 1: Proportional evidence of cycling as a mode of transportation



In fact, it has been observed by the World Health Organization (2020, p. 8) that overall bicycle usage across Europe and the Americas has been on the upward trend in recent years due to various factors including growing cycling infrastructure. The following chart demonstrates on a per-city basis, the change in cyclist participation from the early 1990s to the mid 2010s:

Exhibit 2: Upward historical trend of cyclist participation by city (Americas and Europe)



According to the World Health Organization (2020, p. 3), "every year 41 000 cyclists die in road traffic-related crashes worldwide. Many leave their homes as they would on any normal day – for school, work, worship, or meeting friends – never to return". This statistic is undoubtedly startling, and as mentioned previously, is a main inspiration point for our product.

In addition, moving towards a more Irish-specific background of cyclist safety trends, the Road Safety Authority (2025, p.5) recognised that "between 2020 and 2024, 45 cyclists were killed, and 1,278 cyclists were seriously injured". This is clear evidence that while cyclist road safety is a global issue, Ireland is no exception. Therefore, one could reasonably conclude that Irish consumers who are actively reliant or involved in cycling on a daily basis would potentially stand to benefit from a product that enhances cyclist road safety and accident prevention (See *Appendix A*).

2. IDEA DESCRIPTION

RideAware is driven by the goal of making cycling safer, more intuitive, and more technologically intelligent for everyday cyclists. While existing cycling safety technologies focus on reactive safety and smart glasses models are mostly designed for fitness tracking, communication, our concept focuses on proactive safety and situational awareness. RideAware introduces intelligent cycling glasses that significantly improve situational awareness. These glasses assist cyclists by providing real-time alerts when vehicles approach too closely and by facilitating navigation, all within the rider's immediate visual field. Serving as a personalised safety assistant, the device enhances both the security and confidence of urban cyclists navigating complex traffic environments.

While driving assistant systems (DAS) have been integral to automotive safety since the 1970s (Bengler et al., 2014), similar innovations have rarely been extended to address the specific needs of cyclists. Thus, integrating comparable DAS for cyclists represents a logical and necessary progression in urban mobility safety. Road safety studies show that cyclists are not considered a safe mode of transport due to their high vulnerability compared to motorised vehicles and suffer the most serious consequences in collisions (Evgenikos et al., 2026). Therefore, RideAware addresses one of the biggest barriers that prevents people from cycling: the feeling of being overlooked in traffic or warned too late.

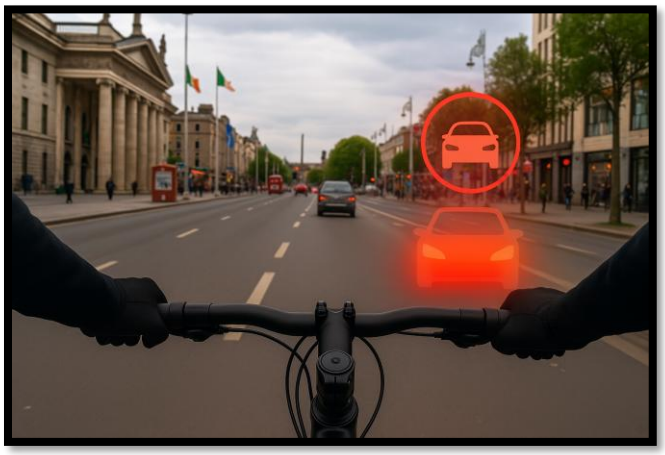
RideAware combines several safety-related functions in a user-centred system as the glasses integrate sensor fusion and GPS technology to provide real-time traffic awareness and collision warnings and contextual, glance-free navigation. The system assists during the day and at night for instance warning cyclists about poorly visible road users such as pedestrians and runners during nighttime. Using an Augmented Reality (AR) Display Module, the system projects AR alerts directly onto the lens, allowing cyclists to maintain full situational awareness without looking away from the road. By combining lightweight hardware and contextual data visualization, this design brings advanced driver-assistance capabilities to cyclists in an accessible, wearable form.

2. IDEA DESCRIPTION

Functionalities

Hazard alert

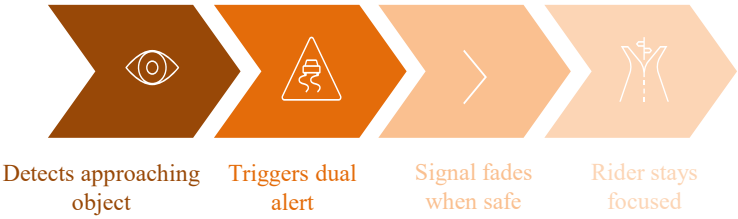
By integrating an Inertial Measurement Unit (IMU), radar and GPS, the system continuously captures motion dynamics such as acceleration, angular velocity, and orientation of the cyclist. A dedicated microcontroller unit (MCU) continuously analyses the surroundings and detects approaching vehicles or other hazards in real time to send out immediate warnings.



Camera Sensor

A rear-facing camera sensor supports object detection and improves hazard awareness by detecting potential danger zones in the cyclist's surroundings, such as approaching vehicles, obstacles on the road or potholes. The system can provide timely visual or audible warnings when bumps or unexpected obstacles are detected, thus enabling cyclists to react early and avoid accidents.

Figure 3: The Hazart Alert Process



Bone conduction

Simultaneously, a haptic warning is transmitted via a bone conduction speaker. By using bone conduction technology, discrete acoustic signals are delivered, leaving the ear canal open and allowing environmental sounds to be heard clearly. A unique, coded tone is played significantly louder on the side (left or right) from which the hazard is approaching. The frequency/cadence of the tone signals the urgency.

The IMU continuously analyses variations in relative motion and orientation to estimate both distance and rate of approach. Once the legal minimum safety distance is breached, the system provides a discreet, low-intensity warning to heighten rider awareness without causing cognitive overload.

The combination of visual and auditory signals in smart glasses for cyclists can improve safety and the perceived sense of safety in road traffic. Studies show that warnings containing both visual and auditory elements are perceived by users as more pleasant and intuitive than purely visual warnings. They also lead to a faster response to dangers (Von Sawitzky, Grauschopf and Riener, 2022). For this reason, both technologies are used. However, all feedback channels (visual, auditory) are designed to integrate naturally into the rider's perception process without distracting the rider from ongoing traffic.

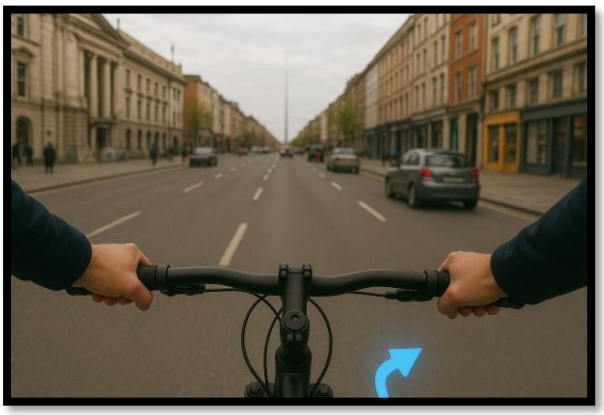
2. IDEA DESCRIPTION

Functionalities

Intelligent AR navigation

A Bluetooth connection links the glasses to the user's smartphone for GPS alert functions.
To ride safely, cyclists have to pay attention to the road ahead without looking down and reduce their speed to prevent collisions when necessary (Chong et al., 2010; Vansteenkiste et al., 2013). Visual attention on the road ahead can also be a challenge when cyclists get distracted by navigation information or smartphone messages (Zhao et al., 2024).

Monocular AR displays enhance hazard awareness by enabling users to detect more dangers during navigation, confirming their potential to improve cyclists' situational perception (Van Lopik et al., 2020). The design must present essential information clearly and concisely and avoid unnecessary complexity that might be distracting to cyclists (Zhao et al., 2024). For this reason, the symbols are only displayed in the fields of vision that do not interfere with the primary field of view.



Using GPS, the glasses offer visual turn-by-turn navigation. Having this information displayed directly in their field of vision enables cyclists to navigate more easily and safely, making both urban and rural journeys smoother and safer (Meng and Tao, 2023). A blue arrow appears at the bottom of the screen and becomes brighter as the next turn approaches. After turning, the display disappears automatically to avoid distraction. This type of minimally invasive AR overlay is particularly effective in human-computer interaction studies, as it provides orientation without distracting attention from the road (Kim & Dey, 2020).

The smart glasses are specially designed for everyday use in urban environments, whether on the way to work or for getting around town. RideAware is designed like regular sports glasses to meet the desire of many cyclists for style and comfort (Lawson et al., 2013).The system is waterproof, making it possible to wear the glasses in any weather condition. The system is powered by a lightweight, rechargeable battery designed for everyday urban cycling.

2. IDEA DESCRIPTION

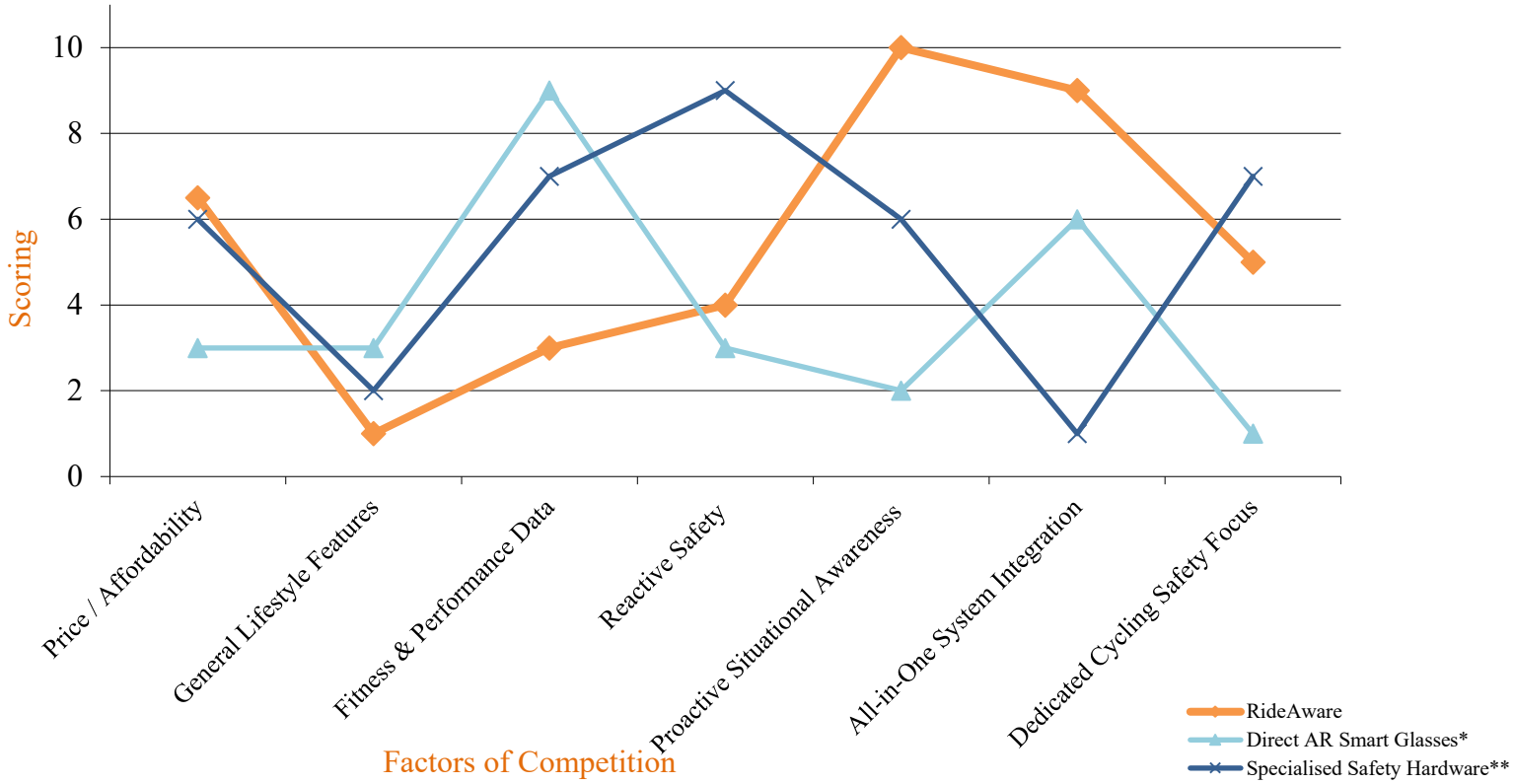
Strategic Positioning

RideAware smart glasses respond to a clear safety gap on the market by fusing multiple technologies into a wearable safety system for cyclists. Current cycling safety technologies offer valuable improvements in visibility and awareness, but they remain primarily reactive in nature. Many smart glasses focused on performance information project visuals directly onto the rider's field of vision using AR. However, RideAware differentiates itself by prioritising active situational awareness and hazard warning within the visual field. RideAware leaps into the blue ocean by introducing a wearable proactive and personalised cycling safety assistance tool.

Cycling safety technology has advanced significantly, offering a range of products designed to enhance rider visibility, awareness, and protection. Specialised safety hardware that include smart helmets, such as the Livall Smart Helmet have Bluetooth speakers, a microphone, an SOS alert system, and LED lighting, allowing riders to make hands-free calls increase their visibility while cycling. Advanced lighting systems further enhance safety, such as the Garmin Varia RTL515 rearview radar taillight, which detects approaching vehicles and alerts the rider.

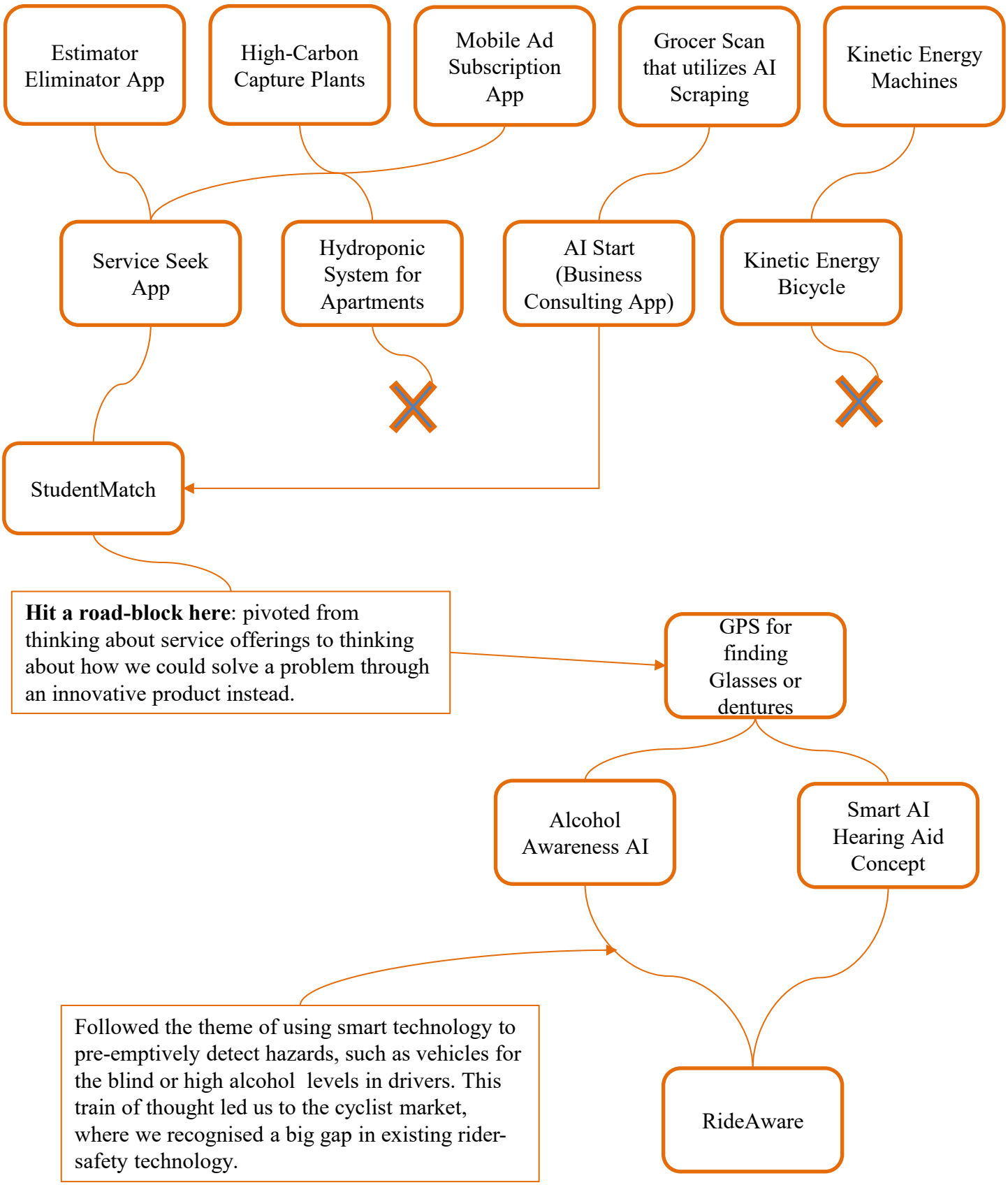
Meanwhile, the main purpose of the Fly6 Pro by Cycliq is to increase passive safety by preserving evidence in the event of accidents or critical incidents in road traffic. Direct AR smart glasses, such as the Oakley Meta Vanguard deliver impressive performance support and are rooted in the realm of reactive feedback and performance tracking. They do not yet offer predictive hazard detection, infrastructure interaction, or proactive risk profiling, which are essential for improving rider safety in dynamic urban environments.

Figure 4: Strategy Canvas of of the Cycling Safety and Wearable Tech Landscape



*ENGO Eyewear, Everysight Maverick, Oakley Meta Vanguard
**Garmin Varia RTL515/RCT715, Livall Smart Helmet, Lumos Ultra Fly Pro, Cycliq Fly6 Pro

3. IDEATION MAP



3. IDEATION MAP

Our starting point was to derive our ideas from actively engaging with global issues and closely observing the small problems of our everyday lives. Ideation is defined as the process of generating, developing and testing ideas that might lead to solutions (Brown, 2008). Our idea generation process was influenced by the principles of the Blue Ocean Strategy (Kim & Mauborgne, 2005) to enter into a new, uncharted marketplace.

Critical questioning was the focus of our selection, but it was equally important to leverage the diverse perspectives of our multinational team to enrich the idea generation and solution finding phase. Our initial idea failed to stand the test when presented to a neutral audience, so we decided to focus on generating a new idea.

Exhibit 5: Four Actions Framework of RideAware

Eliminate

- Distractions (fitness or social media information) and focuses exclusively on safety and navigation
- Warning systems that overwhelm the rider with loud alarms or flashing signals

Raise

- Proactive awareness: accurate and early hazard warnings
- Rider autonomy & control
- Raises vision in all conditions

New Value Curve

Reduce

- Reliance on other traffic participants
- Dependence on smartphones for navigation or alerts
- Weight and bulk
- Reliance on infrastructure

Create

- Proactive safety system for cyclists
- Integrated alerts and navigation in fields of vision
- Connectivity with other smart devices

The process began with each team member independently developing five initial concepts, which were then presented to the group. This was followed by a crucial step: to ensure an unbiased and neutral evaluation. We deliberately introduced a phase of individual reflection, during which each member was able to critically review all concepts independently of the group's influence. Additionally, we engaged with individuals from diverse professional backgrounds, such as engineers to gather external feedback and technical viability insights.

4. MARKET ANALYSIS

Our Customer

A recent press release from RedClick Insurance reports that more than half of the adults in Ireland avoid cycling due to safety concerns (RedClick Insurance, 2025). Particularly, in Dublin, there are several dangerous places for cyclists like the city centre, bridges along the Grand Canal, and further hotspots spread across areas like Rathmines, Ranelagh, Angier Street, and Connolly Station(O'Reilly, 2016). Moreover, serious injuries as well as fatalities among cyclists in Dublin have risen over the past years *(RISA)*. Cyclists feel that motorists are both reckless and careless regarding their presence in the transportation network. Research consistently shows that especially women are significantly more likely than men to avoid cycling due to safety concerns and a lack of confidence in traffic (RedClick, 2025).

Most people who are not cycling are interested but concerned due to the bad bicycle infrastructure. Therefore, the best approach to maximising the population health rewards offered by cycling is to make it safe and convenient (Fishman, 2015). People continue to cycle for a variety of reasons spanning personal health, convenience, mobility needs, and societal benefits, particularly when compared to driving a private car. For instance, physical activity due to cycling resulted in about nine times more gains in life-years than the combined losses in life years due to increased inhaled air pollution doses and traffic accidents (De Hartog et al.,2020).

RideAware offers delivery and courier drivers a compelling Return on Investment (ROI) by solving their critical navigation and safety challenges in new urban areas. Since these drivers constantly use distracting phone navigation to find unfamiliar addresses, RideAware assists by using hands-free, AR-enabled turn-by-turn directions and proactive collision alerts. This not only reduces operational risks and liability costs but also significantly boosts delivery cyclists' efficiency and speed.

Moreover, bicycles are generally increasingly recognised as a functional vehicle for short- to medium-length trips that characterise much of city travel. Furthermore, electric bikes provide effort reduction while still providing most of the health benefits normal cycling does (Fishman, 2015). In light of the advantages, the logical conclusion is to make cycling even more attractive for everyday cyclists by minimising safety concerns and implementing solutions that directly address their primary apprehensions.

4. MARKET ANALYSIS

Potential Customer Personas

Customer Persona 1
Sophie , 32 years old, married
Consultant, Dublin (IRL)
Income: €65,000 per year

Background
Sophie works many hours and does not have a lot of time for exercise. Nevertheless, she wants to stay fit and optimise her time by taking the bike to commute to work daily. She is often annoyed by the heavy traffic and carelessness of other drivers. On the weekend she also rides her bike for fun sometimes.

Needs

Excellent performance at work
Time optimization
Health Optimisation

Pain Points

Chaotic Dublin Traffic
Little time to exercise

Needs

Assistance for perception and agility
Joint friendly exercise
Safe and stress-free exercise

Pain points

Reduced mobility
Awareness of accident risk
Navigation

Customer Persona 2
Giuseppe, 53 years old , Married, 2 children
Doctor
Income: €155,000 per year

Background
Giuseppe is an Orthopaedic surgeon and commutes to work to maintain cardiovascular fitness. With growing age, his mobility and agility have reduced. Moreover, he sees the impact of accidents in his daily work and has a demanding job.

Customer Persona 3
Felicia., 28 years old, single
Physiotherapist
Income: €155 000 per year

Background
In her free time, Felica loves to cycle but she recently had a bike accident and is scared to get back out there. She realized how quickly an accident can happen and wants to ride again with confidence.

Needs

Feeling safe on the road again
Gain back confidence

Pain points

Fear of accidents
Navigation on new routes

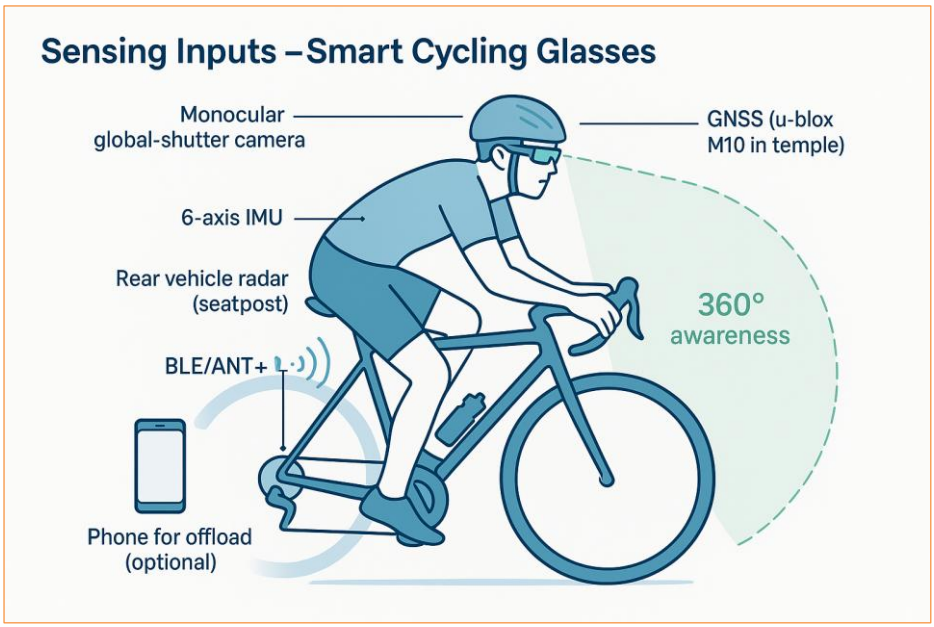
5. TECHNICAL EVALUATION

System Overview:

The RideAware's SLAM technology can be broken up into three segments. The first of these involves the sensing inputs that will communicate real-world data to the computing system within our frames. Following this, the second integral component will involve internal micro-computing that will harness data flow through both front-end and back-end processing. Finally, our system will produce these outputs for the user through integrated AR technology within our frames. The outputs will communicate real-time alerts via monocular displays that utilise waveguide micro-projectors, coupled with directional audio that utilises bone conduction. This will all be wrapped up into a high-quality, durable, plastic frame that is standard across the cycling industry.

An integrated 6-axis IMU supplies high-rate motion data that, fused with vision, yields visual-inertial SLAM for drift-resistant head pose and mapping (Forster et al., 2017; Dellaert & Kaess, 2017; Scaramuzza, 2019). A seat post-mounted rear radar continuously reports the range and closing speed of overtaking vehicles, while telemetry is transmitted to the glasses over BLE/ANT+ for low-latency and low-energy integration. There will also be an Embedded GNSS in the temple, e.g., an ultra-low-power u-blox M10 module, which adds absolute positioning for navigation and geo-anchored warnings (U-blox, 2024). The MCU processor will then synchronise and filter these inputs into unified collision estimations to drive hazard prediction and prioritise alerts.

Exhibit 6: System Overview



Sensing Inputs

RideAware's frames feed a low-power MCU with four sensing streams to deliver 360° awareness. A front monocular, global-shutter camera provides crisp imagery without rolling-shutter skew at cycling speeds (Schubert et al., 2019; Norée Palm, 2023).

This multi-sensor design balances robustness and battery life while covering forward hazards, rear approaches, and location context in a compact, ride-ready package.

5. TECHNICAL EVALUATION

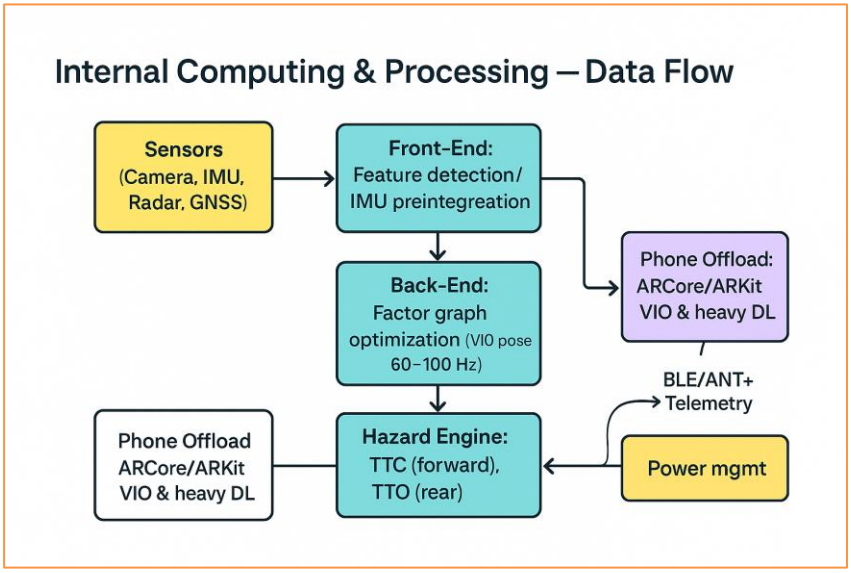
Internal Computing and Processing (Data Flow)

RideAware's computing centers on a low-power MCU that handles sensor I/O, time-sync, BLE, display driving, and power management, enabling real-time assist without bulk (Infineon, 2023). The front-end extracts/tracks visual features and pre-integrates IMU readings; the back-end runs factor-graph optimisation to output high-rate head pose for VIO/SLAM (Forster et al., 2017; Dellaert & Kaess, 2017). The MCU fuses forward video and rear-radar streams to estimate time-to-collision (TTC) for looming hazards and time-to-overtake (TTO) from closing range vehicles, then triggers visual/audio alerts when thresholds are crossed (Lee, 1976). To maximise battery life, heavy VIO/TTC computation is offloaded to the user's smartphone, leveraging existing ARKit/ARCore world-tracking architecture (Apple ARKit; Google ARCore), while the glasses focus on rendering HUD and handling inputs and outputs. Links will be lightweight for optimal telemetry: Radar → Glasses via BLE/ANT+ using existing profiles; Glasses ←→ Phone over BLE 5 with the 2-Mb/s PHY and ~7.5-ms intervals, which is ample for telemetry and draw commands (Nordic Semiconductor, 2024).

Data Outputs (AR Tech):

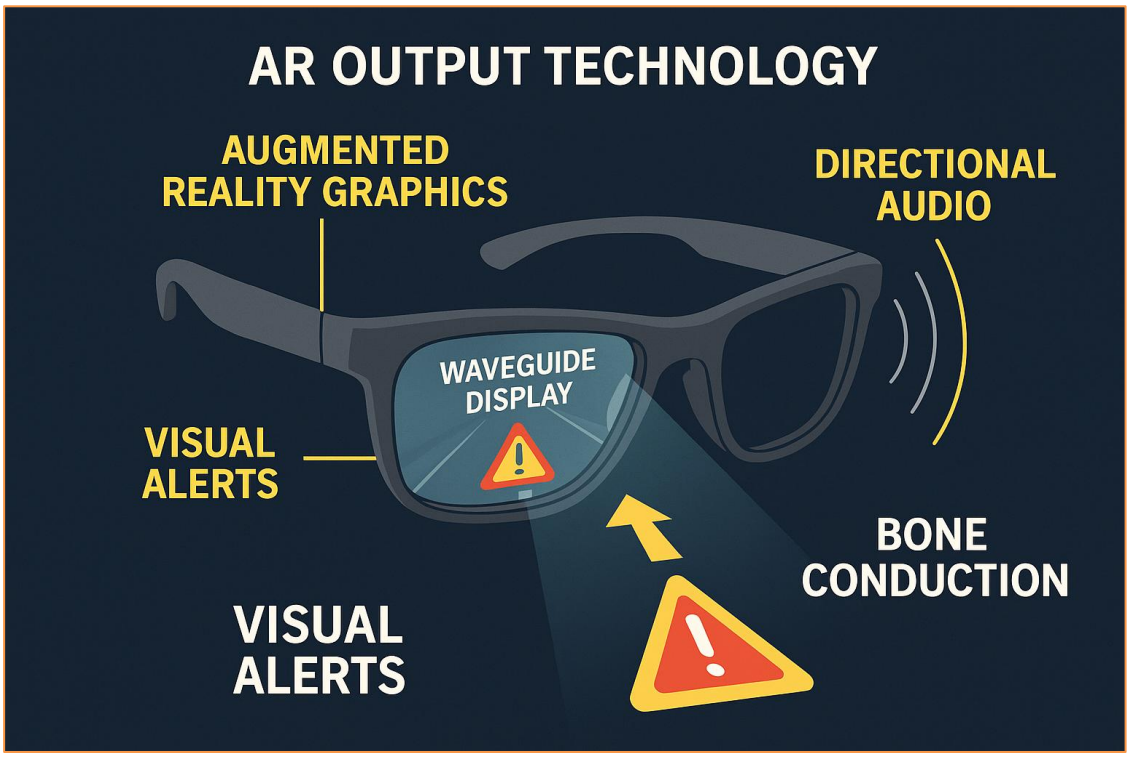
RideAware's output stack (phone app) turns sensed hazards into minimal visual and auditory cues. Visually, a monocular waveguide uses total internal reflection (TIR) with in/out-coupling gratings to expand the eyebox so graphics stay visible as the eye moves and appear at a comfortable virtual distance (Rolland & Goodsell, 2024). The app's HUD compositor converts events into world-locked glyphs by mapping coordinates *world to head* to display from VIO/SLAM, while heavier perception is kept on the phone to extend battery runtimes. To keep overlays realistic, we target motion-to-photon of approximately 20–50 ms, aligning with extended reality comfort measurements (Warburton et al., 2022). Brightness and contrast will auto-adapt via an ambient sensor, and the display will use a high-efficiency LCoS micro-display for outdoor legibility at high resolution (Lee et al., 2018). Audibly, bone-conduction transducers at the temples will deliver directional, coded tones (e.g., left ear = left-side threat), with pitch/cadence scaling with urgency. However, audio levels are capped because excessive bone-conducted audio can reduce environmental awareness (Surendran et al., 2023). Together, these choices will produce high-quality, low-distraction guidance without occluding the real world

Exhibit 7: Internal Computing and Data Flow



5. TECHNICAL EVALUATION

Exhibit 8: AR Output Technology



Power Processing and Hardware:

Accurate AR head-motion estimation requires pairing a global-shutter front camera with a calibrated 6-axis IMU; global shutters avoid rolling-shutter artifacts that degrade visual odometry at higher speeds (Schubert et al., 2018). The monocular waveguide uses a micro-LCoS light engine for daylight-readable, see-through imagery because waveguide combiners are optically inefficient. Therefore, LCoS's robustness and power efficiency make it preferable to OLED for our purposes (Ding et al., 2023). Audio is delivered via temple-mounted bone-conduction transducers that keep the ear canal open and preserve situational awareness (Ellsperman et al., 2021). A u-blox M10 GNSS provides always-on geo-anchoring at approximately 25 mW (u-blox, 2024). Telemetry uses BLE 5 (2M PHY; ~7.5 ms intervals) between phone to glasses and glasses to rear radar— which is ample for HUD updates and IMU streaming (Bluetooth, 2025). With approximately 1.9 Wh cell (500 mAh @ 3.8 V), HUD+Radar (display+GNSS+BLE, camera off) draws approximately 0.2 W for about 9–10 hours of battery life, consistent with ENGO-class sports HUDs (Engo, 2025). AR-perception (camera+VIO+HUD) can approach approximately 1 W, yielding about 1–2 hours of battery life. Therefore, heavier vision processing is offloaded to the phone to extend runtime.

5. TECHNICAL EVALUATION

Product Delivery Model

RideAware will operate under a hybrid model that utilises both a direct-to-consumer approach and in-person retail approach. This strategy will enable us to offer our frames at a lower price while also retaining autonomy on customer experience by providing customers with service touchpoints at retail locations. These retail locations will be a few select bike shops around the city that can help with fitting, calibration, or after-sales support. To ensure there is buy-in on behalf of these specialty retail shops, RideAware will forfeit a marginal portion of profit/unit to receive shelf space at popular bike shops across the city.

Our DTC online sales segment will be run through the RideAware website and fulfillment will utilise an EU 3PL hub to ensure we consistently meet customer expectations when it comes to European shipping times, which currently sit between a 3 to 5-day delivery window (Business wire, 2022). Post-sale, RideAware's companion app will act as an integral component of the customer experience, as this will manage user onboarding, BLE pairing, firmware updates for the glasses, and rider analytics.

Product Design

RideAware will offer a singular SKU: an aviator-inspired frame that combines the practicality of high coverage and concave road-cycling frames, while elevating the style with the timeless look of an aviator. The material that we will be opting for in our frames is Grilamid TR-90, a material that is widely used in performance eyewear due to its durability, low-density, and stress-crack resistance (EMS-Grivory, 2024). These TR-90 frames will be coupled with polycarbonate lenses, a material that provides high-impact resistance and is superior to plastic polymer alternatives (Uvex).

Exhibit 9: Sample Mock-Up of Product Design 1

Exhibit 10: Sample Mock-Up of Product Design 2

Manufacturing Practices

The overall approach to the RideAware manufacturing process involves outsourcing to China in order to maintain cost effectiveness and feasibility regarding pricing and sustainable business practices. More specifically, we would likely base our manufacturing operations in Shenzhen. According to the CAMAL Group (2024), "This region is known for its high concentration of factories, especially in the electronics, machinery, and textiles industries. Shenzhen has earned a reputation as a global hub for electronics manufacturing, with companies like Huawei, BYD, and Foxconn based there". As outlined in the Product Manufacturing/Component Cost Breakdown later in this report, we can reduce overall estimated costs substantially (e.g., labor costs in China average on the lower end compared to more western or local countries) which will make our product more attractive to consumers given affordability alone.

6. INVESTMENT & RETURNS

As with any business startup, a crucial step in the research and development process is the overall potential for profitability. Looking towards the financial perspective and feasibility of RideAware, we can look at key financial measures estimated from essential elements of the implementation process such as prototypes, component parts, manufacturing, marketing, and legal costs.

It is important to note that the following estimates were computed given the seed budget restriction of €300,000 (i.e., some costs were estimated to allow for a more budget friendly/cost efficient approach). Also, we expect variable costs to drop in the long run once we can achieve economies of scale (i.e., ***some variable component costs are inflated because single unit prices are used***). The charts found in *Appendix B* provide a summary for ease of understanding and conciseness.

Retail Selling Price:

Given industry trends in the consumer electronics space, we believe an appropriate markup on our ***Total Estimated Cost per Unit*** is 30% (Naeem, 2024). Therefore, to obtain the retail selling price we can apply a simple cost-plus markup formula (Wall Street Prep, 2024):

Retail Selling Price = *Total Estimated Cost per Unit x (1+ markup)*

= *€148.93 x (1.30)*

= €193.61

= €194 → rounded*

Exhibit 10: Sales Revenue Forecast Model (Year 1-5):

Given the exponential growth nature of many tech startups, we can estimate the expected sales revenue for years 1 through 5 using a simple ***discrete exponential growth regression model*** (Vraciuc, 2012):

$$R_t = R_1(1+r)^{t-1}$$

In the above model, the variables can be set out as follows:

R_t = Expected revenue at time , t

R_1 = Initial expected revenue at year 1

t = year

Given the following assumptions,

R_1 = ***Retail Selling Price x units sold***$_1$

For simplicity, we can make a moderate assumption of 2000 units sold in year 1, given overall market size. Around 100,000 people cycle to work and education in Ireland (Irish Cycle, 2023). Therefore, we can estimate our year 1 sales using an average market penetration rate of 2% (PXP Financial, 2025).

units sold$_1$ = 100,000 x 0.02

units sold$_1$ = 2000

Hence, initial expected sales revenue at year 1 is as follows,

R_1 = €194 x 2000

R_1 = €388,000

We can now calculate the sales revenue forecasts for years 1 through 5 using *scenario analysis* (Brothwell, 2025).

6. INVESTMENT & RETURNS

Sales Revenue Forecast and Scenario Analysis:

Conservative: we can assume a growth rate, *r* , of ***20%***		
Year	*Calculation*	*Estimated Sales Revenue*
1	-	= €388,000
2	= €388,000 (1.20)	= €465,600
3	= €388,000 $(1.20)^2$	= €558,720
4	= €388,000 $(1.20)^3$	= €670,464
5	= €388,000 $(1.20)^4$	= €804,557

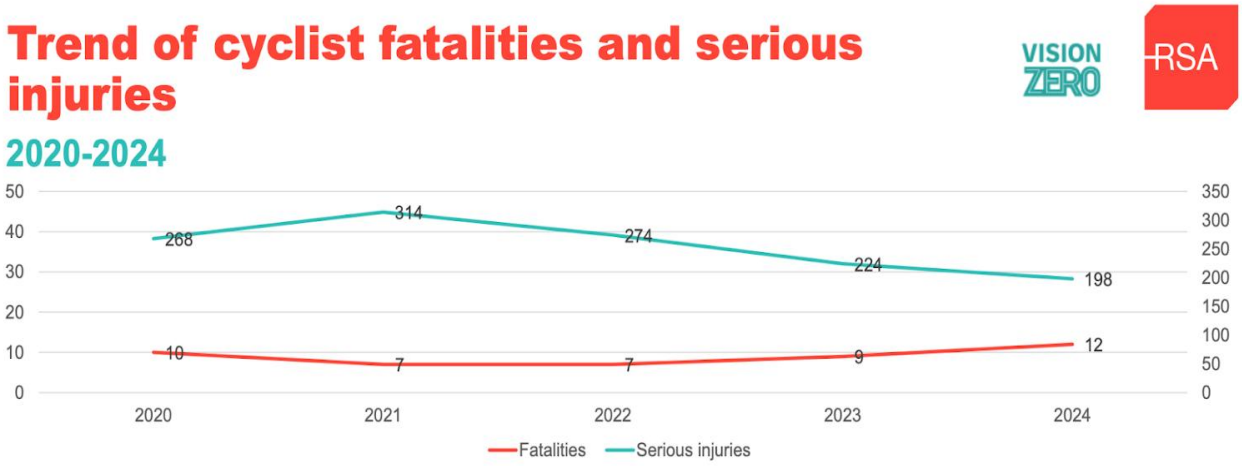
Moderate: we can assume a growth rate, *r* , of ***30%***		
Year	*Calculation*	*Estimated Sales Revenue*
1	-	= €388,000
2	= €388,000 (1.30)	= €504,400
3	= €388,000 $(1.30)^2$	= €655,720
4	= €388,000 $(1.30)^3$	= €852,436
5	= €388,000 $(1.30)^4$	= €1,108,167

Aggressive/optimistic: we can assume a growth rate, *r* , of ***50%***		
Year	*Calculation*	*Estimated Sales Revenue*
1	-	= €388,000
2	= €388,000 (1.50)	= €582,000
3	= €388,000 $(1.50)^2$	= €873,000
4	= €388,000 $(1.50)^3$	= €1,309,500
5	= €388,000 $(1.50)^4$	= €1,964,250

7. APPENDICES

Appendix A:

Recent fatalities and injuries involving cyclists according to the RSA (Ireland Specific):



7. APPENDICES

Appendix B:

Cost Estimates and Structure:

Line Item	*Composition of Line Item*	*Estimated Cost*	*Benchmark Source*
R&D/Prototypes	Engineering and Design Costs	Proof of Concept stage: €10,000 Pilot Sample Stage: €30,000 Final Prototype stage :€100,000	Using an engineering firm who has prior expertise in wearable electronics, we were able to obtain approximate comparable estimates: Here
Product Packaging Design	Packaging Design Costs (low-mid end)	$5000 (€4300 approximately)	Here
Launch Marketing Costs	Marketing Costs (SEO)	€500/month	Here
Website Cost	-	€7000	Here
Initial Injection Mold Cost	-	€2500-€2600	Here
Insurance Costs (Yearly)	Product Liability Insurance Costs	€800-€2000	Here
Trademarking and Product Patent Cost Breakdown:			
Irish Trademark Registration	IPOI Filing Fees and Legal Costs	€1500	1, 2, 3
Trademark Classes (additional)	Potential Filing Costs to Cover Additional Classes	€400	
Irish Patent (national only)	Professional Legal Drafts (Patent Attorney Fees) and IPOI Filing Fees	€6000	1, 2, 3
Renewals	Trademark (10 year) and Patents (annual)	€250, and €300/year on average, respectively	1, 2

7. APPENDICES

Appendix B (Continued):

Product Manufacturing/Component Cost Breakdown:		
AR Display (Composed of Waveguide and Micro OLED Display)	*OLED Display Component:* €56/unit *Waveguide Component:* €15/unit	*OLED Display Component: Here* *Waveguide Component: Here*
Radar Module (rear- mounted)	*€2.45/unit*	*Here*
MCU (Includes built-in compatible BLE and ANT+)	*€5.07/unit*	*Here*
Bone Conduction Speaker	*€9.89/unit*	*Here*
GPS/GNSS and IMU	*GPS Module:* €27/unit *Bosch IMU Module:* €3.50/unit	*GPS Module:* Here *Bosch IMU Module:* Here *+Specifications* Here
Camera (micro-lens)	€3/unit	Here
Battery (Lithium-Ion)	€6.87/unit	Here
Custom Molded Plastic Frames (Injection- Molded)	Assuming initial mid-volume production: €2.24/unit	Here
Wireless Charging Stand (***Note:*** we have used an aftermarket reference here for simplicity as custom design quotes are needed)	€12.05/unit	Here

7. APPENDICES

Appendix B (Continued):

Labour Costs (Outsourced)	Labor Costs: €3.50/unit	Labor Costs: Here	
Packaging and Shipping	Average cost for cardboard packaging (contingent on dimensions): $0.91 – $ 12.86/unit (approximately €0.78 – €11.07/unit) → due to the nature of our product size, we can assume this number will be on the lower end of the range (e.g., assume €2 for total per unit calculation) Shipping (China-IE): *Assumptions:* 1.Air freight $4.20 (approximately €3.62) per kg, 2. and per unit product weight </= 100g. €0.36/unit	Average cost for Cardboard packaging: Here Shipping (China-IE): Here	
Total Estimated Cost per Unit:		Calculated as the sum of *Manufacturing/ Component Costs:* = ***€148.93***	

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DECLARATION OF AUTHORSHIP

Assignment Submission

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Programme: Bachelor of Arts in Global Business
Project Title: New Enterprise Development Project
Module Code: BAA1003
Lecturer: Prof Eric Clinton
Project Due Date: 13/10/2025

Declaration

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We have not copied or paraphrased an extract of any length from any source without identifying the source and using quotation marks as appropriate. Any images, audio recordings, video or other materials have likewise been originated and produced by us or are fully acknowledged and identified.

This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study. We have read and understood the referencing guidelines found at https://www.dcu.ie/library/citing-referencing and/or recommended in the assignment guidelines.

We understand that we may be required to discuss with the module lecturer/s the contents of this submission.

We/us/our incorporates we/us/our in the case of group work, which is signed by all of us.